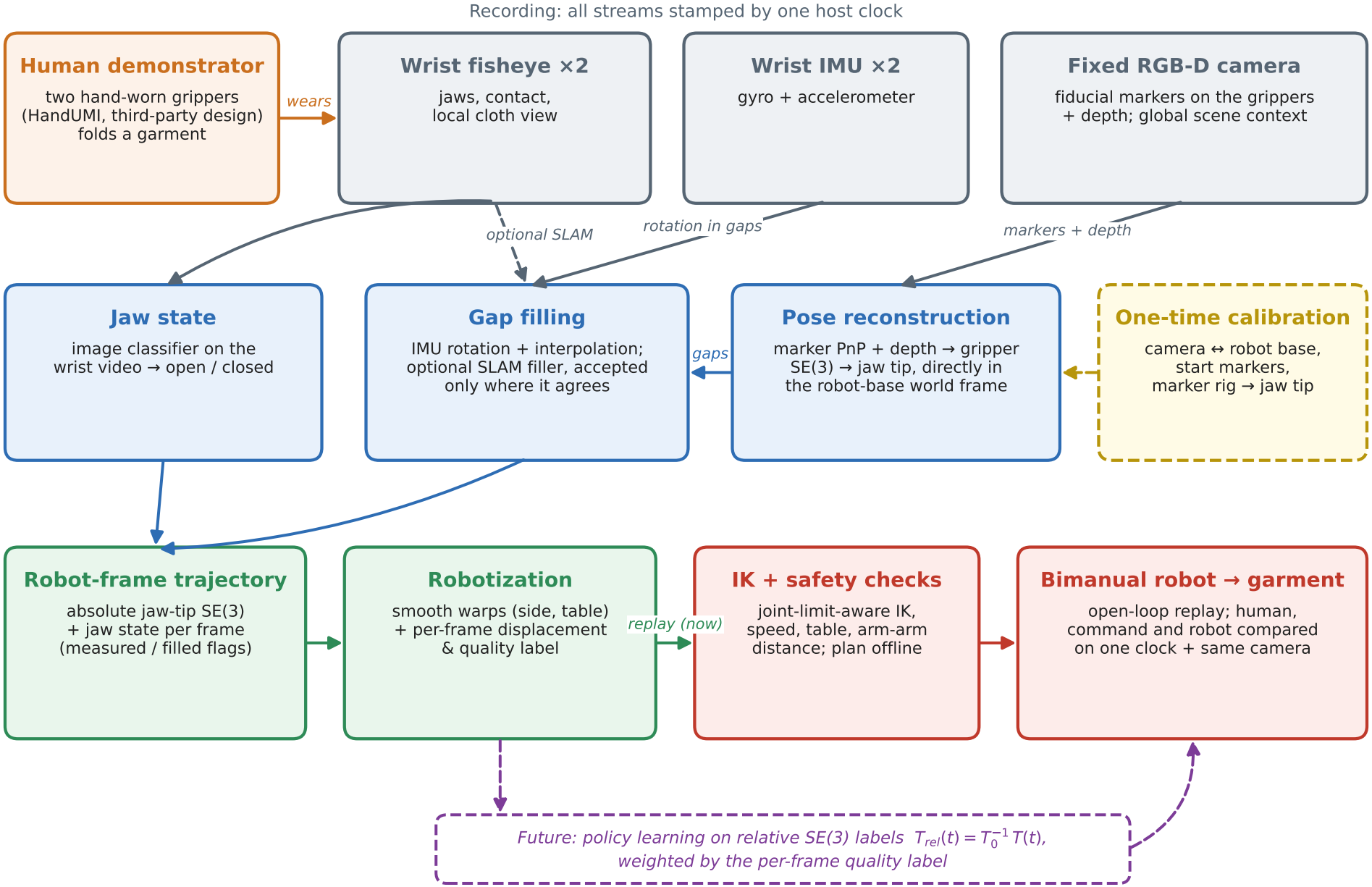
Recording: all streams stamped by one host clock
Human demonstrator
two hand-worn grippers (HandUMI, third-party design) folds a garment
wears
Wrist fisheye ×2
jaws, contact, local cloth view
Wrist IMU ×2
gyro + accelerometer
Fixed RGB-D camera
fiducial markers on the grippers + depth; global scene context
optional SLAM
rotation in gaps
markers + depth
Jaw state
image classifier on the wrist video → open / closed
Gap filling
IMU rotation + interpolation; optional SLAM filler, accepted only where it agrees
gaps
Pose reconstruction
marker PnP + depth → gripper SE(3) → jaw tip, directly in the robot-base world frame
One-time calibration
camera ↔ robot base, start markers, marker rig → jaw tip
Robot-frame trajectory
absolute jaw-tip SE(3) + jaw state per frame (measured / filled flags)
Robotization
smooth warps (side, table) + per-frame displacement & quality label
replay (now)
IK + safety checks
joint-limit-aware IK, speed, table, arm-arm distance; plan offline
Bimanual robot → garment
open-loop replay; human, command and robot compared on one clock + same camera
Future: policy learning on relative SE(3) labels $T_{rel}(t) = T_0^{-1} T(t)$, weighted by the per-frame quality label